Paper 78 — Deep Residual Learning

Kaiming He, Xiangyu Zhang, Shaoqing Ren, Jian Sun (CVPR 2016)

Deep Residual Learning for Image Recognition

Ref 108 | A / D; F / B | Must read | Original paper

The paper in one minute

Question: Why can adding layers make a neural network *harder* to train, even when the deeper model can represent the shallower one?
Model/Method: Replace direct learning of a mapping $H(x)$ by learning its residual $F(x) = H(x) - x$; construct identity-shortcut residual blocks and test very deep CNNs.
Finding: Residual networks optimize much more successfully than comparable plain networks as depth increases; the authors train models up to 152 layers on ImageNet and explore 100/1000-layer CIFAR models.
Why we care: Architecture can alter the difficulty of optimization without changing the intended prediction task. Residual connections are an empirical design breakthrough, not a universal global-convergence theorem.

1. What is the degradation problem?

If a shallow network fits the training data, a deeper network of sufficient capacity *could* simply copy the shallow mapping and make added layers act as identities. Yet the authors observe that plain networks become *harder to optimize*: a 56-layer plain model has *higher training error* than a 20-layer counterpart (Figure 1). This is not merely overfitting, because the training error itself is worse.

2. The residual reformulation

Instead of forcing stacked layers to directly learn $H(x)$, let them learn

$$F(x; W) = H(x) - x, \qquad y = F(x; W) + x.$$

If the desirable mapping is already close to the identity, the residual function can be close to zero. A shortcut copies x directly to the output and does not itself introduce trainable parameters in the identity case.

Worked scalar example

Imagine the desired transformation is $H(x) = 1.05x$. A plain layer would learn multiplication by 1.05; the residual branch only needs to learn

$$F(x) = 0.05x, \qquad y = x + 0.05x = 1.05x.$$

If $F(x) = 0$, the block immediately implements the identity. This toy example illustrates the representational convenience; it is not a general proof that optimization must be fast.

3. Why shortcuts can improve backpropagation

For a residual block, its input-output Jacobian is

$$\frac{\partial y}{\partial x} = I + J_F(x).$$

For a scalar residual branch $F(x) = 0.05x$, the Jacobian is 1.05 rather than 0.05. Gradients have a direct identity contribution through each shortcut, even when the learned branch is small. This helps explain why residual pathways can provide more direct signal propagation; it does not imply that vanishing or exploding gradients become impossible in arbitrary deep ResNets.

A subtle but crucial distinction
The authors' central observation is about *optimization difficulty*, as shown by training error, not merely test accuracy or theoretical expressivity.

4. What exactly is implemented?

The basic residual unit applies a stack of convolutions/normalizations and adds the shortcut: $y = F(x; W) + x$. If feature dimensions differ, the paper discusses zero padding or a learned projection shortcut, usually written $W_s x$; shortcut and residual outputs must have the same shape.

Plain versus residual comparison

Experiments control architecture families so that a plain network and a corresponding residual network have similar layered components, but the residual one adds shortcuts. Training curves show that very deep plain networks can have worse training error, whereas residual counterparts benefit substantially from depth.

Bottleneck blocks for very deep nets

The ImageNet 50/101/152-layer designs use a bottleneck pattern with 1×1, 3×3, and 1×1 convolutions, reducing and then restoring channel width. This lowers computational cost relative to stacking wide 3×3 convolutions at every stage.

5. How to interpret the experiments

CIFAR-10. The paper shows successful optimization of residual networks with 100 and even 1000 layers, illustrating that very deep architectures are trainable in this design. **ImageNet.** A 152-layer residual network achieves a 4.49% single-model top-5 *validation* error, while an ensemble attains 3.57% top-5 *test* error in ILSVRC 2015. *Do not attribute the ensemble score to a single model.* **Transfer.** Strong image-recognition representations also improve COCO object detection, supporting the practical importance of depth but not isolating every causal mechanism of residual optimization.

Connection to the survey
This is a foundational *empirical architecture paper*. Later results on deep ResNet optimization and depth-dependent conditioning analyze more restrictive mathematical models; those later theorems should not be attributed retroactively to He et al.

6. Contributions, assumptions, limitations

Contributions. (i) Residual learning through identity shortcuts; (ii) a controlled demonstration of depth-related *training-error degradation* in plain nets; (iii) practical very-deep ResNet designs; (iv) substantial ImageNet and transfer results. **Assumptions/context.** Residual additions require aligned tensor dimensions; projected shortcuts may be used when shape changes. Empirical comparisons use particular initialization, batch normalization, SGD schedules, datasets, and CNN block designs. **Limitations.** The work does not prove global optimality, absence of spurious minima, or saddle escape. The Jacobian intuition does not constitute an exact trajectory theorem for 152-layer SGD. Residual and plain architectures also can differ in optimization statistics and parameterization.

7. Cross-paper connections

Du et al. (2019): provides convergence theory for specified overparameterized ResNet settings. **Nguyen (2021):** studies convergence with favorable feature conditioning. **Saxe (2014):** explains nonlinear optimization dynamics in deep *linear* nets. Here, the architectural shortcut acts as an intervention on trainability, not an added saddle-escape perturbation.

8. Quick reference

Residual block	$y = x + F(x; W)$ (identity shortcut).
Target reformulation	$F(x) = H(x) - x$.
Gradient route	$\partial y/\partial x = I + J_F$.
Projection shortcut	$y = F(x) + W_s x$ when shapes require alignment.
Observed obstruction	Plain-network training error can worsen with depth.
ImageNet outcome	152-layer single: 4.49% validation top-5; ensemble: 3.57% test top-5.
Theorem status	Strong empirical evidence; no universal global-GD theorem.

Papers Master note

He et al. show that learning residual functions using identity shortcuts greatly improves the optimization of very deep CNNs, overcoming training-error degradation observed in plain networks. Important architectural foundation for later ResNet convergence theory, not a saddle-escape or global-optimality proof.

Original: https://www.cv-foundation.org/openaccess/content_cvpr_2016/papers/He_Deep_Residual_Learning_CVPR_2016_paper.pdf.